

What are the technical specifications of your saucepan?

As more and more electronics gets embedded into everyday objects, there arises a need to understand systems, software and networks yet again

"Electronics has become the medium through which innovation is being pushed through into various consumer markets. The payload—or the key value-add—of the increased electronics content is the embedded software and the specialised hardware that the new electronics deliver. Further, these software and hardware designs employ complex mathematical algorithms, signal-processing capabilities, sensor integration and other functions. Consequently, they are able to bring new and innovative capabilities in applications that previously have been mundane and non-hi-tech," explains Dr Arun Mulpur, manager-Industry Marketing, MathWorks Inc. "The primary impact of Electronics of Things is the manifold increase in the capabilities, functionality, intelligence and usefulness of products, devices and services in many industries and applications. However, this certainly comes at a cost—including the increase in complexity of system design and integration, verification, validation and risk of failure."

Smart watches, intelligent ice cubes and socially-active T-shirts might sound very simple and fascinating, but the fact is that they are not easy to build. One needs to understand the systems, software and protocols involved in the development of such devices that rub shoulders with human beings day in and day out. Fortunately, almost every major player in the IT and electronics industry is developing platforms and solutions for the IoT, making it easier to understand and implement electronics in common objects. Here is a quick peek into the world of 'everyday' electronics!

A story of systems and platforms. Dr Mulpur says that in this age, "It is no longer exclusively software or hardware. System on chip (SoC) designs have ARM or other embedded software platforms as a part of the chip fabric. So, the engineers working on such designs must be ready to leverage these new capabilities and the design tools and the methodologies that support these engineers must evolve as well."

"SoCs at scales unheard of are being made possible by deep innovations in the semiconductor value chain—design, EDA and manufacturing. Apart from creating immense integration and miniaturisation opportunities, lower technology nodes have also heralded an era of extremely power-efficient solutions. Networking enables devices to communicate and make them meaningful in settings that were thought to be out of bounds. The effect is anybody's guess," says Vipin N., senior director-Advanced Technology, Ricoh Innovations Pvt Ltd.

There are several chips and SoCs available in the market today. Freescale offers ARM-based QorIQ communications processors, the Kinetis 32-bit ARM MCU and the i.MX 32-bit ARM MPU for the electronics/IoT. Intel has strengthened its embedded portfolio with

Atom and with the recent addition of the Quark range of products, and also has communication products in the form of cellular modems, Wi-Fi chips and a range of software products from Wind River.

Open source solutions are also becoming popular in the electronics of things age, because embedding electronics into everyday objects requires everybody to be a 'maker'! Arduino and Raspberry Pi, are two popular open source platforms for such DIY projects. "I think that Arduino as a tool is very empowering, not only because of its robustness, but also because of its prompt customer support," says Priya Kuber, managing director, Arduino-India.

Smarter software. "Embedded software is typically designed keeping in mind more constraints like available memory, response time, etc, compared to general-purpose software," says S. Natarajan, senior program manager-India Strategy & Planning, Intel.

Vivek Tyagi, country sales director, Freescale Semiconductor, adds that software developed for such embedded electronics have to work on low power and also invariably consume low energy. Such solutions are designed in a manner to perform multiple tasks in a very short span of time.

"Software developers of embedded electronics go that extra yard to ensure security. Power efficiency is another issue that gets a rather fair share of attention. Short product innovation times coupled with short 'windows of opportunities' also necessitate that a bulk of the processing be done by the firmware. The new breed of embedded electronics has also resulted in unique architectures and novel solutions from different schools of thought—be it the use of instructions sets defined by software or open CL for massive data processing. An example is the implementation of a software-defined radio where a bulk of the modem is implemented in the software!" explains Vipin.

Shivesh Vishwanathan, senior consultant, (Mobility)—Technology Consulting Group, Persistent Systems, shares his company's perspective: "We call the software developed for embedded electronics, consumer goods and devices as consumerware, and there are ten specific characteristics of consumerware that differentiate it from traditional software written for desktop computers or the web. The first amongst the top three characteristics of consumerware is that it is service oriented as opposed to being packaged, which means that business models can be service based as well. Second is that it is very contextual to the user's needs, preferences, location, presence, etc. So, it is very fresh and different for each user, and if your app looks and behaves the same for any two users, you need to revisit what you are doing. The third is that consumerware is used very socially—even by the 'things' in the IoT

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What caused the sudden surge in electronics usage in everyday objects?

The range of electronics-embedded devices is simply unbelievable, but what has caused this sudden surge in the use of electronics in everyday objects?

Price and innovations in hardware. Shivesh Vishwanathan, senior consultant (Mobility)—Technology Consulting Group, Persistent Systems, explains, "There are two primary rea-

sons for the increasing amount of electronics in day-to-day products. First, the price of electronics components has come down significantly and second, the innovations that are making hardware components smaller, more malleable, etc. These advances are really capturing people's imagination and nothing seems to be out of bounds. The resulting effect is that the line between what we call computer and what we call everyday products is blurring. The software and electronics industries

increasingly resemble the fashion or consumer goods industry with a lot of brands, consumer preferences playing pivotal roles, trends changing very fast, etc."

Availability and programmability. Anand reasons further that, "About a decade ago, electronics was not as pervasive as it is today. Most of the electronics items were analogue and not very programmable. This was one of the impediments in making electronics more interesting and experimental. However,